

Solution Architecture

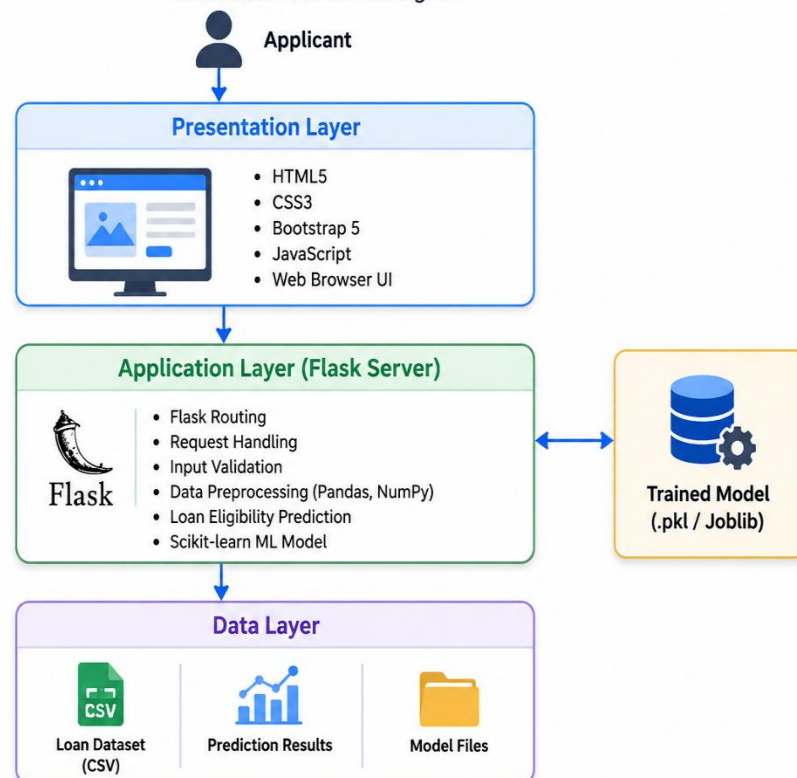
Date	05 July 2026
Team ID	SL-2026-02
Project Name	Smart Lender – Loan Eligibility Prediction Using Machine Learning
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.

Smart Lender – Loan Eligibility Prediction Using Machine Learning

Solution Architecture Diagram



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides a responsive web interface where users enter applicant details and view loan eligibility results.	HTML5, CSS3, Bootstrap, JavaScript
API Gateway / Application Layer	Flask handles HTTP requests, validates user inputs, communicates with the prediction model, and returns the eligibility result.	Python, Flask
Core Logic Service	Performs data preprocessing, feature transformation, loads the trained machine learning model, and predicts loan eligibility.	Scikit-learn, Pandas, NumPy, Joblib
Data / Storage Layer	Stores the loan dataset used for training and the serialized machine learning model for deployment.	CSV Dataset, Joblib/Pickle Files